# Capstone Project: Chronic Kidney Disease (CKD) Prediction

## 1. Problem Statement

Chronic Kidney Disease (CKD) is a serious global health issue that often goes undetected until it reaches advanced stages. Early diagnosis is critical to prevent complications, reduce healthcare costs, and improve patient outcomes.

However, diagnosing CKD typically requires analyzing multiple clinical and laboratory indicators, which can be time-consuming and prone to human error.

**This project aims to develop a machine learning model that can accurately predict the presence and the stage of CKD using patient medical data, enabling early detection and supporting clinical decision-making.**

## 2. Dataset Source

The dataset is from Kaggle and it contains patient health indicators related to kidney function. It includes variables in categories of

- **Demographics, Vital Signs, Laboratory Tests, Urine Analysis**, **Medical History**

Two datasets are provided:

- **Training dataset:** Includes features + target variable (CKD status)
- **Test dataset:** Also includes target variable (used for validation/evaluation in this case)

## 3. Proposed Approach / Modeling Techniques

A. Data Preprocessing

B. Exploratory Data Analysis (EDA)

C. Feature Engineering

D. Modeling Techniques.

- Perform Traditional methods such as Logistic Regression ,Decision Trees ,Random Forest Gradient Boosting (XGBoost),

- Perform a  Deep Learning modelling -Artificial Neural Networks (ANN) using Keras

E. Model Evaluation

- Use metric such as **Classification (Binary):** Accuracy, Precision, Recall , F1-score, ROC-AUC

F. Model Optimization

- Hyperparameter tuning (Grid Search / Random Search)
- Cross-validation
- Avoid overfitting (Dropout, Regularization)

# 4. Expected Outcomes

- A **high-performing predictive model** for CKD detection
- Identification of **key risk factors** (e.g., creatinine, blood pressure)
- Improved understanding of **CKD progression patterns**
- A system that can:
    - Predict CKD presence and the stage